Online Appendix for "Financial Incentives and Performance: A Meta-Analysis of Experiments in Economics"*

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Abstract

Economists typically model financial incentives as enhancing performance, whereas psychologists emphasize that incentives can backfire. Experimental findings are mixed. We collect 2,193 estimates from 88 economics experiments and account for 48 contextual factors. Using recent advances in correcting for publication bias and p-hacking, we find that the corrected mean effect of financial incentives on performance is close to zero across most field contexts. Laboratory settings and loss framing yield statistically significant but modest positive effects even after bias correction. Our results suggest that increasing financial rewards rarely produces large performance gains in the experimental settings most studied by economists.

Keywords: Incentives, experiments, meta-analysis, model uncertainty, publication bias

JEL Codes: C90, D91, M52

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Additional Statistics and Results

Figure B1: No systematic differences in results across countries

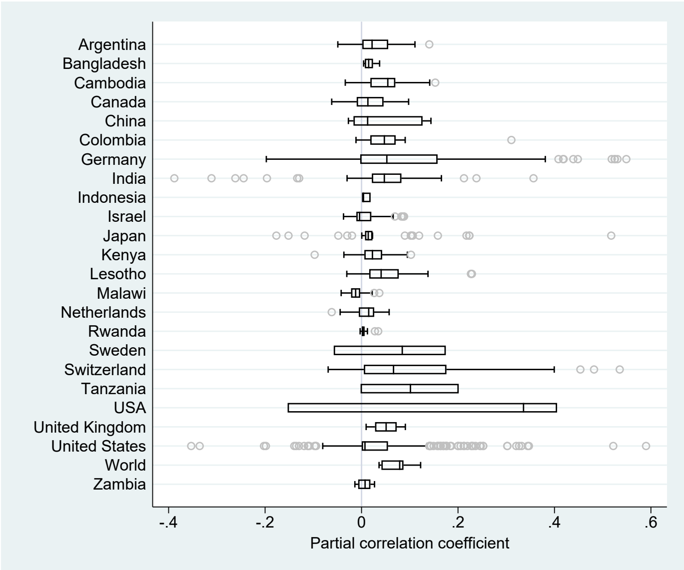
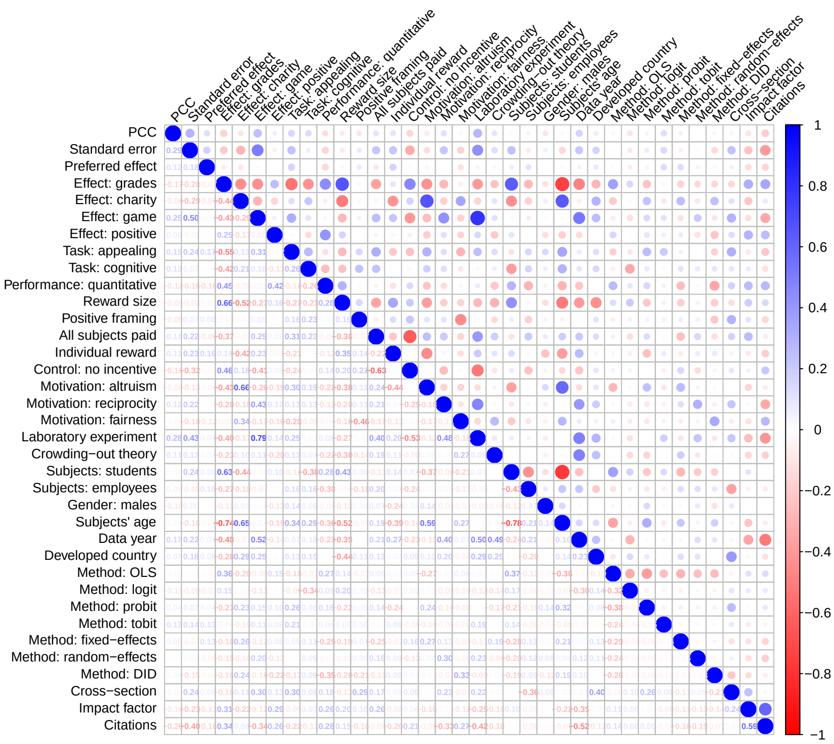


Figure B2: Correlation matrix for regression variables



Note: The figure shows correlation coefficients for regression variables. Blue color (dark in grayscale) indicates positive correlation, while red color (light in grayscale) indicates negative correlation.

Related meta-analyses Our analysis is most closely related to two strands of meta-analytic research: meta-analyses of financial incentives in psychology (Weibel *et al.*, 2010; Kim *et al.*, 2022), and recent meta-analyses in experimental economics (Brown *et al.*, 2024; Matousek *et al.*, 2022). Weibel *et al.* (2010) include 46 primary studies, one estimate per study, covering both economics and psychology experiments, though only 11 of their studies (24%) are from economics. Their main finding is that financial incentives can reduce performance in interesting tasks, consistent with the motivation crowding theory. However, their analysis does not correct for publication bias or model uncertainty, and their classification of task characteristics is not standardized across fields. Kim *et al.* (2022) build on this literature using 69 primary studies and 82 effect sizes. Their sample is based predominantly on laboratory settings in psychology, and their study likewise does not apply formal corrections for publication bias. Only three of the studies in their sample overlap with ours, indicating minimal direct overlap with the economics literature.

Brown *et al.* (2024) and Matousek *et al.* (2022) represent recent meta-analyses in experimental economics. Brown *et al.* (2024) study loss aversion using 150 primary studies and 607 effect sizes, documenting substantial publication bias and highlighting the importance of heterogeneity in framing and study design. Matousek *et al.* (2022) analyze the individual discount rate based on 59 studies and 927 estimates, again finding considerable methodological heterogeneity and evidence of publication bias. While these studies focus on different behavioral parameters, their methodological approach is similar to ours: employing meta-regression, model averaging, and bias correction techniques. Our contribution differs by focusing specifically on financial incentives and performance outcomes, covering 88 primary studies and 2,193 estimates, and introducing a broader set of contextual moderators and robustness checks.

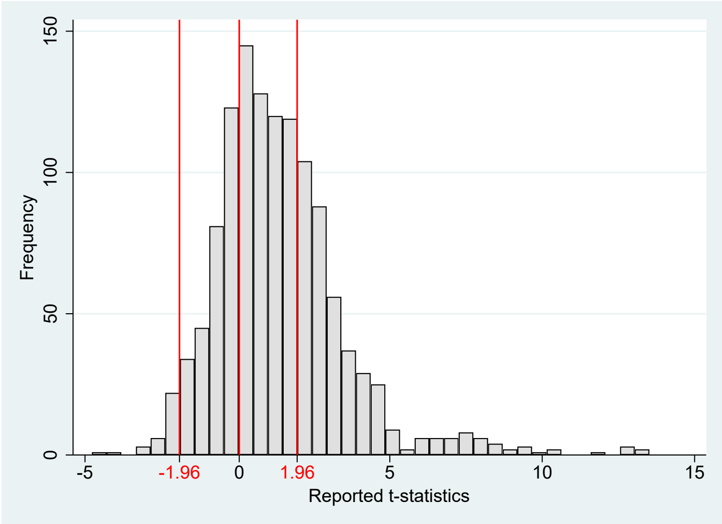
Multiple estimates per study For 93% of the studies we collect multiple estimates, with an average of 25 estimates included per study. This large number is primarily due to the fact that most primary studies report estimates from multiple model specifications. Additionally, many studies incorporate several treatment arms. For example, Fershtman & Gneezy (2011), Dwenger *et al.* (2016), and Levitt *et al.* (2016) explore different incentive levels, resulting in at least one estimate per treatment condition. Many studies conduct subsample analyses, as in Dohmen & Falk (2011), Duflo *et al.* (2012), and Karlan & List (2007). Additionally, the authors of primary studies frequently employ diverse estimation frameworks. For example, Karlan & List (2007) use both OLS and Probit models; Angrist *et al.* (2009) apply OLS and TSLS; Englmaier *et al.* (2024) employ OLS, Probit, and Cox regression; and Carrillo *et al.* (2021) utilize both OLS and difference-in-differences estimation. Many studies incorporate additional robustness checks, such as Ito (2015), Meier (2007), Sliwka & Werner (2017), Kremer et al. (2009), and Lazear (2000). Another reason for collecting multiple estimates is that some studies include multiple experiments within a single paper: for example, Charness & Gneezy (2009), Ariely *et al.* (2009), Gneezy & Rustichini (2000), and Gneezy & List (2006), among others.

Motivation crowding-out models in economics The integration of economic theory with the concept of motivation crowding-out, originally developed in psychology, provides a nuanced framework for understanding how external incentives may undermine intrinsic motivation. One of the earliest and most influential empirical contributions in economics is by Frey & Oberholzer-Gee (1997), who show that offering monetary compensation for hosting a nuclear waste facility reduces individuals' willingness to accept it. Their findings suggest that financial incentives can crowd out civic-minded behavior, establishing an empirical foundation for the exploration of motivation crowding-out within economic contexts.

Building on this, Benabou & Tirole (2003) develop a formal model in which agents derive utility from both intrinsic and extrinsic rewards. They show that the introduction of monetary incentives can act as a signal that the activity is unpleasant or unimportant, thereby undermining intrinsic motivation. In subsequent work, Tirole & Benabou (2006) examine how incentives affect prosocial behavior, and Benabou & Tirole (2011) extend the analysis to include identity and moral values, showing how these interact with external incentives in shaping behavior. Another important contribution comes from Sliwka (2007), who embeds motivation crowding into a principal-agent model with endogenous trust. His model explores how different levels of control and monitoring influence the agent's motivation, and under which conditions trust is more effective than incentives.

Taken together, these models show that crowding-out is not merely a psychological concept but has been rigorously formalized in economic theory. They emphasize the importance of considering unintended consequences of extrinsic rewards, particularly when intrinsic motivation plays a key role.

Figure B3: The distribution of t-statistics peaks at zero



Notes: The figure represents the distribution of t-statistics corresponding to the effect of financial incentives on performance reported in the literature. Vertical lines represent zero and critical values associated with statistical significance at the 5% level.

Tests based on the distribution of t-statistics and p-values Here we present additional approaches to modeling publication bias that do not rely on the uncorrelation assumption. These only test for the bias and do not yield an estimate of the corrected mean effect. Because the models use the reported t-statistics (or p-values), the results cannot be affected by the normalization to partial correlation coefficients that we choose to ensure compatibility in the case of all the previous techniques. The first additional approach is the so-called caliper test due to Gerber & Malhotra (2008). The caliper test focuses on an important threshold of the t-statistic (typically 1.96, which denotes statistical significance at the 5% level, or 0, which denotes a change in sign) and compares the number of reported t-statistics just below and just above the threshold. In the absence of publication bias and with a sufficiently narrow caliper, there should be no difference. Additionally, Elliott *et al.* (2022) derive two new rigorously founded techniques that do not require us to define the location of the thresholds. The techniques rely on the conditional chi-squared test of Cox & Shi (2023). The first technique is a histogram-based test for nonincreasingness of the p-curve, the second technique is a histogram-based test for 2-monotonicity and bounds on the p-curve and the first two derivatives.

Figure B3 shows the distribution of reported t-statistics in the case of the literature on incentives and performance. The histogram suggests a jump at zero but not at the thresholds associated with 5% significance, suggesting that publication bias or p-hacking arises mainly due to selection for intuitive sign, not for statistical significance. Panel A of Table B1 shows the results of the caliper test and corroborates the patterns seen in the histogram. No jump in the distribution of t-statistics is related to statistical significance at the 5% level. For the threshold of zero, in contrast, we observe a jump when narrow calipers are considered. As expected, the evidence for a jump disappears when we widen the caliper. Panel B shows the results of p-hacking tests due to Elliott *et al.* (2022). The main advantage of these rigorous tests is that they do not need us to specify a threshold of the t-statistic: they test for possible bias using the general distribution of all p-values. When all estimates, preferred or non-preferred, are considered, we reject the null hypothesis of no p-hacking/publication bias for all our samples: baseline, extended, and working papers. When we restrict attention to preferred estimates, only the test for monotonicity and bounds rejects the null hypothesis at the 5% level. This is not surprising because with preferred estimates we work with a much smaller sample (even after considering all estimates with p-values below 0.15, not just 0.1 as in the previous case), and the tests of Elliott *et al.* (2022), in particular the test for non-increasingness, are known to have relatively low power (Havranek *et al.*, 2024; Elliott *et al.*, 2024).

Table B1: Tests based on the distribution of t-statistics and p-values

Panel A: Caliper tests due to Gerber & Malhotra (2008)			
Threshold for t*-statistic:* -1.96	caliper: 0.05	caliper: 0.10	caliper: 0.15
Share above threshold minus 0.5	0.333 (0.167)	0.136 (0.152)	0.088 (0.123)
Observations	6	11	17
Threshold for t*-statistic:* $+1.96$	caliper: 0.05	caliper: 0.10	caliper: 0.15
Share above threshold minus 0.5	0.119 (0.109)	-0.035 (0.077)	-0.041 (0.058)
Observations	21	43	74
Threshold for t*-statistic:* 0	caliper: 0.05	caliper: 0.10	caliper: 0.15
Share above threshold minus 0.5	0.292^{***} (0.085)	0.160^{**} (0.066)	0.071 (0.054)
Observations	24	53	84
Panel B: Tests due to Elliott *et al.* (2022)			
All estimates	Baseline	Expanded	Working papers
Test for non-increasingness	0.020	0.001	0.000
Test for monotonicity and bounds	0.000	0.000	0.000
Observations ($p <= 0.10$)	547	809	241

Continued on next page

Table B1: Tests based on the distribution of t-statistics and p-values (continued)

Total observations	1,252	1,785	408
Preferred estimates	Baseline	Expanded	Working papers
Test for non-increasingness	0.075	0.077	0.114
Test for monotonicity and bounds	0.008	0.002	0.000
Observations ($p <= 0.15$)	243	351	115
Total observations	396	567	157

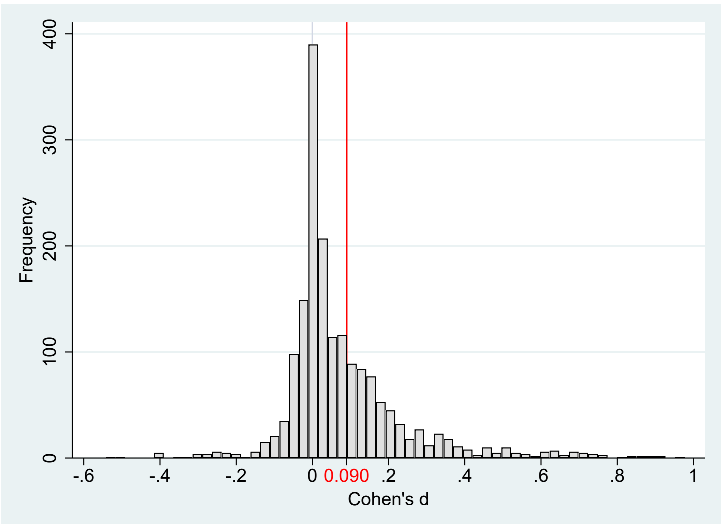
Notes: In Panel A, the tests compare for the baseline dataset the relative frequency of estimates above and below an important threshold for the t-statistic. A test statistic of 0.292, for example, means that 79.2% estimates are above the threshold and 20.8% estimates are below the threshold. Standard errors are clustered at the study level. Panel B reports for different subsamples the p-values of two tests developed by Elliott *et al.* (2022), which also feature cluster-robust variance estimators (null hypothesis: no p-hacking or publication bias). ** $p < 0.05$, *** $p < 0.01$.

Table B2: Summary statistics for Cohen's d (expanded dataset and working papers)

		Unweighted			Weighted		
	No. of obs.	Mean	95% conf. int.		Mean	95% conf. int.	
Top 50 journals							
All estimates	1,785	0.090	0.081	0.098	0.146	0.134	0.157
Preferred estimates	567	0.121	0.103	0.140	0.189	0.165	0.212
Working papers							
All estimates	408	0.108	0.083	0.134	0.098	0.072	0.124
Preferred estimates	157	0.130	0.087	0.174	0.098	0.055	0.141

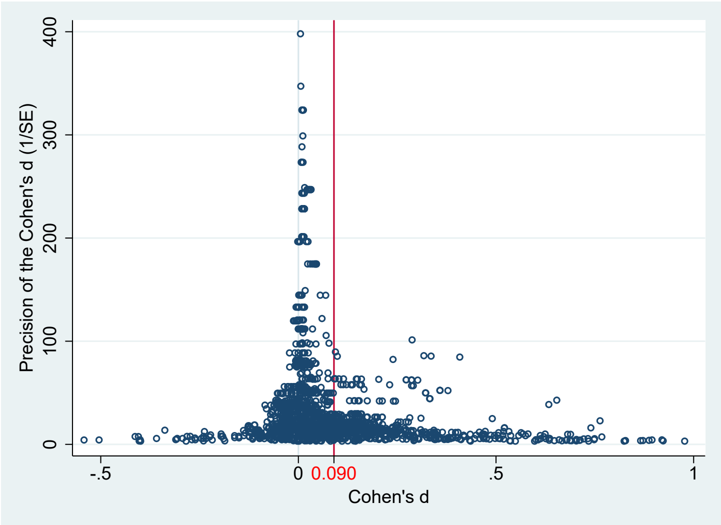
Notes: The table summarizes Cohen's d coefficients corresponding to the estimated effects of financial incentives on performance reported in individual studies. Preferred estimates are those emphasized by the authors of individual studies.

Figure B4: Histogram for the expanded dataset



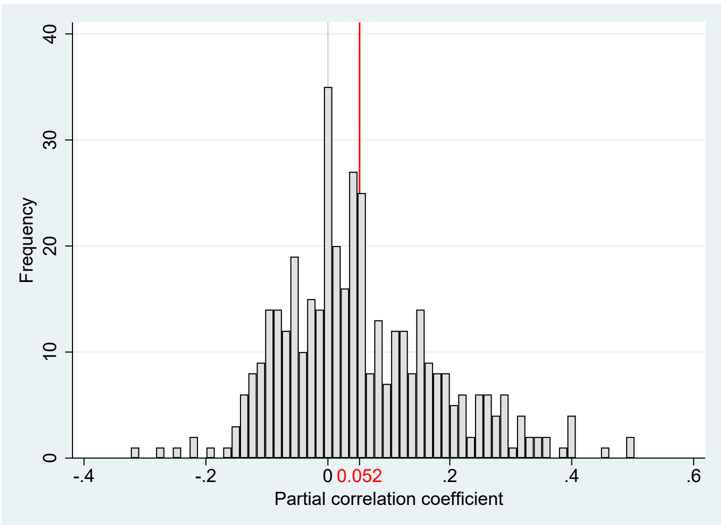
Notes: The figure depicts a histogram of Cohen's d coefficients corresponding to the estimated effects of financial incentives on performance reported in individual studies. The vertical line denotes the sample mean. Outliers are excluded from the figure for ease of exposition but included in all statistical tests.

Figure B5: Funnel plot for the expanded dataset



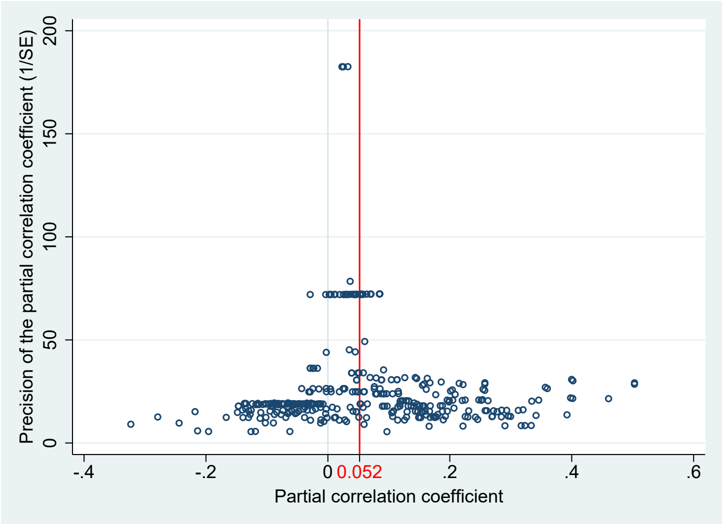
Notes: The figure shows Cohen's d coefficients corresponding to the estimated effects of financial incentives on performance reported in individual studies. The vertical line denotes the sample mean. In the absence of publication bias, the funnel should be symmetrical around the most precise estimates, and the mean should align with those most precise estimates.

Figure B6: Histogram for working papers



Notes: The figure depicts a histogram of the partial correlation coefficients corresponding to the estimated effects of financial incentives on performance reported in individual working papers. The vertical line denotes the sample mean. Outliers are excluded from the figure for ease of exposition but included in all statistical tests.

Figure B7: Funnel plot for working papers



Notes: See notes to Figure B5.

Table B3: Publication bias tests for the baseline dataset (Cohen's d)

Panel A: Linear techniques					
	OLS	FE	BE	Study	Precision
Publication bias	1.065***	1.458***	0.713***	0.889***	1.294***
(Standard error)	(0.273)	(0.511)	(0.246)	(0.335)	(0.304)
Effect beyond bias	0.0229	-0.00753	0.0690**	0.0500*	0.0132
(Constant)	(0.0195)	(0.0395)	(0.0335)	(0.0266)	(0.0119)
Observations	1,252	1,252	1,252	1,252	1,252
Panel B: Nonlinear techniques					
	Top10	WAAP	Stem	AK	Kink
Publication bias				P = 0.298	1.294***
				(0.0588)	(0.131)
Effect beyond bias	0.024***	0.0227***	0.0255	0.003	0.0132***
	(0.00572)	(0.00701)	(0.0483)	(0.00235)	(0.00306)
Observations	1,252	1,252	1,252	1,252	1,252
Panel C: Endogeneity-robust techniques					
				MAIVE	p-uniform*
Publication bias				0.400***	L = 177.2
				(0.120)	(p < *0.001*)
				{0.388, 1.346}	
Effect beyond bias				0.0200**	0.133***
				(0.00954)	(0.00625)
First-stage robust *F-stat*				2,701	
Observations				1,252	1,252

Notes: Panel A: Results of regression $D_{is} = D_0 + \gamma SE(D_{is}) + \epsilon_{is}$, where D_{is} denotes Cohen's d coefficient corresponding to the i-th estimate from the s-th study and $SE(D_{is})$ denotes its standard error. The standard errors of the regression parameters are clustered at the study level and shown in parentheses. OLS = ordinary least squares, FE = study fixed effects, BE = study between effects, Study = weighted by the inverse of the number of estimates reported per study, Precision = weighted by the inverse of the estimate's standard error. Panel B: WAAP = weighted average of adequately powered estimates (Ioannidis *et al.*, 2017). Top10 = the method due to Stanley *et al.* (2010) focusing on the most precise estimates. Stem = the stem-based method due to Furukawa (2020). Kink model = the endogenous kink method due to Bom & Rachinger (2019). AK = the selection model due to Andrews & Kasy (2019), where P denotes the probability that estimates insignificant at the 5% level are published relative to the probability that significant estimates are published (the latter normalized at 1). Panel C: MAIVE = a linear version of the meta-analysis instrumental variable estimator due to Irsova *et al.* (2025), in which we use the inverse of the square root of the number of observations as an instrument for the standard error. In curly brackets we show the Anderson-Rubin 95% confidence interval. P-uniform* = the method by van Aert & van Assen (2025), where L denotes the statistic of the publication bias test; the corresponding p-value is in parenthesis (null hypothesis: no bias). *** and ** denote statistical significance at the 1% and 5% level.

Table B4: Publication bias tests for the expanded dataset (Cohen's d)

Panel A: Linear techniques					
	OLS	FE	BE	Study	Precision
Publication bias	1.141***	1.405***	0.968***	1.084***	1.538***
(Standard error)	(0.242)	(0.426)	(0.241)	(0.323)	(0.316)
Effect beyond bias	0.0126	-0.00525	0.0493	0.0377	0.00178
(Constant)	(0.0154)	(0.0288)	(0.0308)	(0.0211)	(0.00126)
Observations	1,785	1,785	1,785	1,785	1,785
Panel B: Nonlinear techniques					
	Top10	WAAP	Stem	AK	Kink
Publication bias				P = 0.278	1.538***
				(0.0888)	(0.082)
Effect beyond bias	0.0102***	0.00266***	0.00782	0.000879	0.00178***
	(0.00097)	(0.00055)	(0.0183)	(0.00165)	(0.000514)
Observations	1,785	1,785	1,785	1,785	1,785
Panel C: Endogeneity-robust techniques					
				MAIVE	p-uniform*
Publication bias				0.950***	L = 2.261
				(0.226)	(p = *0.133*)
				{0.524, 1.376}	
Effect beyond bias				0.0255	0.00820***
				(0.0157)	(0.00117)
First-stage robust *F-stat*				3,100	
Observations				1,785	1,785

Notes: Panel A: Results of regression $D_{is} = D_0 + \gamma SE(D_{is}) + \epsilon_{is}$, where D_{is} denotes Cohen's d coefficient corresponding to the i-th estimate from the s-th study and $SE(D_{is})$ denotes its standard error. The standard errors of the regression parameters are clustered at the study level and shown in parentheses. OLS = ordinary least squares, FE = study fixed effects, BE = study between effects, Study = weighted by the inverse of the number of estimates reported per study, Precision = weighted by the inverse of the estimate's standard error. Panel B: WAAP = weighted average of adequately powered estimates (Ioannidis *et al.*, 2017). Top10 = the method due to Stanley *et al.* (2010) focusing on the most precise estimates. Stem = the stem-based method due to Furukawa (2020). Kink model = the endogenous kink method due to Bom & Rachinger (2019). AK = the selection model due to Andrews & Kasy (2019), where P denotes the probability that estimates insignificant at the 5% level are published relative to the probability that significant estimates are published (the latter normalized at 1). Panel C: MAIVE = a linear version of the meta-analysis instrumental variable estimator due to Irsova *et al.* (2025), in which we use the inverse of the square root of the number of observations as an instrument for the standard error. In curly brackets we show the Anderson-Rubin 95% confidence interval. P-uniform* = the method by van Aert & van Assen (2025), where L denotes the statistic of the publication bias test; the corresponding p-value is in parenthesis (null hypothesis: no bias). *** and ** denote statistical significance at the 1% and 5% level.

Table B5: Publication bias tests for the expanded dataset (partial correlation coefficients)

Panel A: Linear techniques					
	OLS	FE	BE	Study	Precision
Publication bias	0.931***	0.407	0.731***	0.696***	1.624***
(Standard error)	(0.226)	(0.359)	(0.265)	(0.251)	(0.330)
Effect beyond bias	0.0134*	0.0303**	0.0359**	0.0375***	0.000852
(Constant)	(0.00778)	(0.0116)	(0.0154)	(0.0102)	(0.000653)
Observations	1,785	1,785	1,785	1,785	1,785
Panel B: Nonlinear techniques					
	Top10	WAAP	Stem	AK	Kink
Publication bias				P = 0.286	1.607***
				(0.0887)	(0.087)
Effect beyond bias	0.0051***	0.00134***	0.00391	0.0005	0.00085***
	(0.00048)	(0.000278)	(0.00985)	(0.0005)	(0.000273)
Observations	1,785	1,785	1,785	1,785	1,785
Panel C: Endogeneity-robust techniques					
				MAIVE	p-uniform*
Publication bias				0.973***	L = 187.1
				(0.241)	(p = *0.001*)
				{0.520, 1.427}	
Effect beyond bias				0.0120	0.0675***
				(0.00794)	(0.00306)
First-stage robust *F-stat*				4,549	
Observations				1,785	1,785

Notes: Panel A: Results of regression $PCC_{is} = PCC_0 + \gamma SE(PCC_{is}) + \epsilon_{is}$, where PCC_{is} denotes the partial correlation coefficient of the i-th estimate from the s-th study and $SE(PCC_{is})$ denotes its standard error. The standard errors of the regression parameters are clustered at the study level and shown in parentheses. OLS = ordinary least squares, FE = study fixed effects, BE = study between effects, Study = weighted by the inverse of the number of estimates reported per study, Precision = weighted by the inverse of the estimate's standard error. Panel B: WAAP = weighted average of adequately powered estimates (Ioannidis *et al.*, 2017). Top10 = the method due to Stanley *et al.* (2010) focusing on the most precise estimates. Stem = the stem-based method due to Furukawa (2020). Kink model = the endogenous kink method due to Bom & Rachinger (2019). AK = the selection model due to Andrews & Kasy (2019), where P denotes the probability that estimates insignificant at the 5% level are published relative to the probability that significant estimates are published (the latter normalized at 1). Panel C: MAIVE = a linear version of the meta-analysis instrumental variable estimator due to Irsova *et al.* (2025), in which we use the inverse of the square root of the number of observations as an instrument for the standard error. In curly brackets we show the Anderson-Rubin 95% confidence interval. P-uniform* = the method by van Aert & van Assen (2025), where L denotes the statistic of the publication bias test; the corresponding p-value is in parenthesis (null hypothesis: no bias). *** and ** denote statistical significance at the 1% and 5% level.

Table B6: Publication bias tests for the baseline dataset (no winsorizing)

Panel A: Linear techniques					
	OLS	FE	BE	Study	Precision
Publication bias	0.776***	0.127	0.509	0.458	1.379***
(Standard error)	(0.282)	(0.431)	(0.309)	(0.359)	(0.323)
Effect beyond bias	0.0225**	0.0465***	0.0434**	0.0460***	0.00658
(Constant)	(0.0103)	(0.0160)	(0.0193)	(0.0133)	(0.00586)
Observations	1,252	1,252	1,252	1,252	1,252
Panel B: Nonlinear techniques					
	Top10	WAAP	Stem	AK	Kink
Publication bias				P = 0.31	1.382***
				(0.0521)	(0.145)
Effect beyond bias	0.0123***	0.0102***	0.0393	0.0013	0.00657***
	(0.00282)	(0.00292)	(0.0253)	(0.0009)	(0.00163)
Observations	1,252	1,252	1,252	1,252	1,252
Panel C: Endogeneity-robust techniques					
				MAIVE	p-uniform*
Publication bias				0.869***	L = 187.4
				(0.307)	(*p = 0.001*)
				{0.292, 1.446}	
Effect beyond bias				0.0190	0.0675***
				(0.0106)	(0.00306)
First-stage robust *F-stat*				13,087	
Observations				1,252	1,252

Notes: Panel A: Results of regression $PCC_{is} = PCC_0 + \gamma SE(PCC_{is}) + \epsilon_{is}$, where PCC_{is} denotes the partial correlation coefficient of the i-th estimate from the s-th study and $SE(PCC_{is})$ denotes its standard error. No winsorizing is applied. The standard errors of the regression parameters are clustered at the study level and shown in parentheses. OLS = ordinary least squares, FE = study fixed effects, BE = study between effects, Study = weighted by the inverse of the number of estimates reported per study, Precision = weighted by the inverse of the estimate's standard error. Panel B: WAAP = weighted average of adequately powered estimates (Ioannidis *et al.*, 2017). Top10 = the method due to Stanley *et al.* (2010) focusing on the most precise estimates. Stem = the stem-based method due to Furukawa (2020). Kink model = the endogenous kink method due to Bom & Rachinger (2019). AK = the selection model due to Andrews & Kasy (2019), where P denotes the probability that estimates insignificant at the 5% level are published relative to the probability that significant estimates are published (the latter normalized at 1). Panel C: MAIVE = a linear version of the meta-analysis instrumental variable estimator due to Irsova *et al.* (2025), in which we use the inverse of the square root of the number of observations as an instrument for the standard error. In curly brackets we show the Anderson-Rubin 95% confidence interval. P-uniform* = the method by van Aert & van Assen (2025), where L denotes the statistic of the publication bias test; the corresponding p-value is in parenthesis (null hypothesis: no bias). *** and ** denote statistical significance at the 1% and 5% level.

Table B7: Publication bias tests for the expanded dataset (no winsorizing)

Panel A: Linear techniques					
	OLS	FE	BE	Study	Precision
Publication bias	1.266***	1.788**	1.059***	1.290**	1.554***
(Standard error)	(0.325)	(0.695)	(0.326)	(0.518)	(0.318)
Effect beyond bias	0.00576	-0.0299	0.0469	0.0236	0.00169
(Constant)	(0.0191)	(0.0475)	(0.0427)	(0.0339)	(0.00124)
Observations	1,785	1,785	1,785	1,785	1,785
Panel B: Nonlinear techniques					
	Top10	WAAP	Stem	AK	Kink
Publication bias				P = 0.280	1.554***
				(0.0532)	(0.083)
Effect beyond bias	0.0102***	0.00233***	0.00782	0.000878	0.00169***
	(0.00097)	(0.000554)	(0.0183)	(0.00101)	(0.000513)
Observations	1,785	1,785	1,785	1,785	1,785
Panel C: Endogeneity-robust techniques					
				MAIVE	p-uniform*
Publication bias				0.962***	L = 2.242
				(0.264)	(*p = 0.134*)
				{0.466, 1.458}	
Effect beyond bias				0.0266	0.00820***
				(0.0169)	(0.00117)
First-stage robust *F-stat*				1,768	
Observations				1,785	1,785

Notes: Panel A: Results of regression $D_{is} = D_0 + \gamma SE(D_{is}) + \epsilon_{is}$, where D_{is} denotes Cohen's d coefficient corresponding to the i-th estimate from the s-th study and $SE(D_{is})$ denotes its standard error. No winsorizing is applied. The standard errors of the regression parameters are clustered at the study level and shown in parentheses. OLS = ordinary least squares, FE = study fixed effects, BE = study between effects, Study = weighted by the inverse of the number of estimates reported per study, Precision = weighted by the inverse of the estimate's standard error. Panel B: WAAP = weighted average of adequately powered estimates (Ioannidis *et al.*, 2017). Top10 = the method due to Stanley *et al.* (2010) focusing on the most precise estimates. Stem = the stem-based method due to Furukawa (2020). Kink model = the endogenous kink method due to Bom & Rachinger (2019). AK = the selection model due to Andrews & Kasy (2019), where P denotes the probability that estimates insignificant at the 5% level are published relative to the probability that significant estimates are published (the latter normalized at 1). Panel C: MAIVE = a linear version of the meta-analysis instrumental variable estimator due to Irsova *et al.* (2025), in which we use the inverse of the square root of the number of observations as an instrument for the standard error. In curly brackets we show the Anderson-Rubin 95% confidence interval. P-uniform* = the method by van Aert & van Assen (2025), where L denotes the statistic of the publication bias test; the corresponding p-value is in parenthesis (null hypothesis: no bias). *** and ** denote statistical significance at the 1% and 5% level.

Table B8: Publication bias tests accounting for interdependence

Panel A: All estimates					
	OLS	W Study	BE Study	W Experiment	BE Experiment
Publication bias	0.974***	0.672**	0.546**	0.782***	0.695***
(Standard error)	(0.284)	(0.333)	(0.241)	(0.300)	(0.225)
Effect beyond bias	0.0211	0.0660***	0.0792**	0.0543**	0.0638**
(Constant)	(0.0147)	(0.0242)	(0.0317)	(0.0244)	(0.0308)
Observations	2,193	2,193	2,193	2,193	2,193
Panel B: Published estimates					
	OLS	W Study	BE Study	W Experiment	BE Experiment
Publication bias	1.141***	1.084***	0.968***	1.163***	1.085***
(Standard error)	(0.242)	(0.323)	(0.241)	(0.288)	(0.227)
Effect beyond bias	0.0126	0.0377	0.0493	0.0270	0.0353
(Constant)	(0.0154)	(0.0211)	(0.0308)	(0.0221)	(0.0305)
Observations	1,785	1,785	1,785	1,785	1,785
Panel C: Preferred published estimates					
	OLS	W Study	BE Study	W Experiment	BE Experiment
Publication bias	1.156***	1.070**	0.923***	1.101**	0.991***
(Standard error)	(0.328)	(0.419)	(0.304)	(0.454)	(0.288)
Effect beyond bias	0.0296	0.0641**	0.0780**	0.0566	0.0653
(Constant)	(0.0247)	(0.0266)	(0.0370)	(0.0401)	(0.0375)
Observations	567	567	567	567	567

Notes: The table presents the results of regression $D_{is} = D_0 + \gamma SE(D_{is}) + \epsilon_{is}$, where D_{is} denotes Cohen's d coefficient corresponding to the i-th estimate from the s-th study and $SE(D_{is})$ denotes its standard error. Standard errors are clustered at the study level and shown in parentheses. OLS = ordinary least squares, W Study = weighted by the inverse of the number of estimates reported per study, BE Study = study between effects, W Experiment = weighted by the inverse of the number of estimates reported per experiment, BE Experiment = experiment between effects. There can be more than one experiment conducted within one study. Panel A also includes working papers (88 studies in total). Panel B only includes published papers (69 studies). Panel C only shows results for estimates emphasized by the authors of primary studies. *** and ** denote statistical significance at the 1% and 5% level.

Table B9: Publication bias tests for working papers

Panel A: Linear techniques					
	OLS	FE	BE	Study	Precision
Publication bias	0.0154	0.361**	0.108	0.0958	1.556***
(Standard error)	(0.210)	(0.167)	(1.002)	(0.410)	(0.173)
Effect beyond bias	0.0513***	0.0346***	0.0420	0.0426	-0.00251**
(Constant)	(0.0120)	(0.00864)	(0.0589)	(0.0313)	(0.00101)
Observations	408	408	408	408	408
Panel B: Nonlinear techniques					
	Top10	WAAP	Stem	AK	Kink
Publication bias				P = 0.0841	1.608***
				(0.0369)	(0.147)
Effect beyond bias	0.0121***	NA	-0.0682	-0.00135	-0.00257***
	(0.00315)	(NA)	(0.0786)	(0.00217)	(0.000856)
Observations	408	408	408	408	408
Panel C: Endogeneity-robust techniques					
				IV	p-uniform*
Publication bias				0.0670	L = 6.018
				(0.789)	(p = *0.014*)
				{-1.260, 1.706}	
Effect beyond bias				0.0488	0.0133***
				(0.0283)	(0.00515)
First-stage robust *F-stat*				423	
Observations				408	408

Notes: Panel A: Results of regression $PCC_{is} = PCC_0 + \gamma SE(PCC_{is}) + \epsilon_{is}$, where PCC_{is} denotes the partial correlation coefficient of the i-th estimate from the s-th study and $SE(PCC_{is})$ denotes its standard error. Only estimates from working papers are considered. The standard errors of the regression parameters are clustered at the study level and shown in parentheses. OLS = ordinary least squares, FE = study fixed effects, BE = study between effects, Study = weighted by the inverse of the number of estimates reported per study, Precision = weighted by the inverse of the estimate's standard error. Panel B: WAAP = weighted average of adequately powered estimates (Ioannidis *et al.*, 2017). Top10 = the method due to Stanley *et al.* (2010) focusing on the most precise estimates. Stem = the stem-based method due to Furukawa (2020). Kink model = the endogenous kink method due to Bom & Rachinger (2019). AK = the selection model due to Andrews & Kasy (2019), where P denotes the probability that estimates insignificant at the 5% level are published relative to the probability that significant estimates are published (the latter normalized at 1). Panel C: MAIVE = a linear version of the meta-analysis instrumental variable estimator due to Irsova *et al.* (2025), in which we use the inverse of the square root of the number of observations as an instrument for the standard error. In curly brackets we show the Anderson-Rubin 95% confidence interval. P-uniform* = the method by van Aert & van Assen (2025), where L denotes the statistic of the publication bias test; the corresponding p-value is in parenthesis (null hypothesis: no bias). *** and ** denote statistical significance at the 1% and 5% level.

Table B10: Specification test for the Andrews & Kasy (2019) model

	Baseline	Baseline preferred
Correlation	0.647 [0.567, 0.668]	0.458 [0.352, 0.564]
Observations	1,252	396
	Expanded	Expanded preferred
Correlation	0.711 [0.699, 0.756]	0.680 [0.597, 0.73]
Observations	1,785	567
	WP	WP preferred
Correlation	0.680 [0.542, 0.728]	0.696 [0.525, 0.779]
Observations	408	157

Notes: Following Kranz & Putz (2022), the table shows, for various subsets of the literature, the correlation coefficient between the logarithm of the absolute value of the estimated inverse elasticity and the logarithm of the corresponding standard error, weighted by the inverse publication probability estimated by the Andrews & Kasy (2019) model. If the assumptions of the model hold, the correlation is zero. Bootstrapped 95% confidence interval in parentheses. Preferred estimates are those emphasized by the authors of individual studies.

Table B11: Economic significance of key variables

	One-std.-dev. change		Maximum change	
	Effect on PCC	% of mean	Effect on PCC	% of mean
Standard error (pub. bias)	0.022	44%	0.097	191%
Laboratory experiment	0.021	41%	0.052	103%
Cross-section	-0.015	-29%	-0.029	-58%
Task: cognitive	0.014	28%	0.033	65%
Positive framing	-0.011	-22%	-0.039	-76%
Gender: males	-0.008	-16%	-0.035	-69%
Task: appealing	0.007	13%	0.013	27%
Control: no incentive	0.007	13%	0.014	28%

Notes: The table presents the marginal influence of selected variables on the partial correlation coefficient (PCC) corresponding to the effect of financial incentives on performance. The column “one-std.-dev. change” shows how the PCC changes when we increase the value of the variable by one standard deviation. The column “maximum change” represents the change in the PCC when the variable is increased from its minimum to its maximum. The percentage values indicate the magnitude of the implied effect in relation to the sample mean (0.051).

Table B12: Robustness checks for Bayesian model averaging

Response variable: Partial correlation coefficient	Bayesian model averaging (alternative priors)			Bayesian model averaging (weights)		
	P. mean	P. SD	PIP	P. mean	P. SD	PIP
Constant	-0.035	NA	1.000	-0.699	NA	1.000
Standard error (pub. bias)	0.687	0.100	1.000	0.503	0.091	1.000
Definition of the performance effect						
Effect: grades	-0.001	0.005	0.033	0.000	0.001	0.009
Effect: charity	0.000	0.002	0.012	0.000	0.002	0.013
Effect: game	0.000	0.001	0.008	0.000	0.001	0.008
Effect: positive	0.013	0.016	0.461	0.007	0.012	0.303
Nature of the task						
Task: appealing	0.013	0.011	0.643	0.025	0.007	0.984
Task: cognitive	0.033	0.008	0.994	0.033	0.007	0.999
Performance: quantitative	-0.007	0.011	0.311	0.000	0.001	0.008
Reward scheme						
Reward size	0.000	0.002	0.018	0.000	0.001	0.015
Positive framing	-0.039	0.013	0.957	-0.012	0.015	0.414
All subjects paid	0.000	0.002	0.022	0.000	0.001	0.009
Individual reward	0.000	0.002	0.016	0.000	0.001	0.011
Control: no incentive	0.014	0.013	0.593	0.000	0.001	0.010
Motivation beyond money						
Motivation: altruism	-0.001	0.004	0.066	-0.001	0.004	0.057
Motivation: reciprocity	0.000	0.001	0.009	0.000	0.001	0.008
Motivation: fairness	0.000	0.003	0.017	0.002	0.006	0.073
Study design						
Laboratory experiment	0.052	0.012	1.000	0.054	0.008	1.000
Crowding-out theory	0.000	0.001	0.013	0.001	0.003	0.052
Structural variation						
Subjects: students	0.000	0.003	0.017	0.000	0.001	0.013
Subjects: employees	0.000	0.002	0.017	0.000	0.001	0.008
Gender: males	-0.035	0.018	0.858	-0.049	0.012	0.995
Subjects' age	0.000	0.002	0.026	0.000	0.001	0.012
Data year	0.008	0.144	0.009	0.096	0.008	1.000
Developed country	0.000	0.001	0.007	0.000	0.002	0.019
Estimation technique						
Method: OLS	0.000	0.002	0.030	0.000	0.002	0.025
Method: logit	0.000	0.001	0.007	0.001	0.006	0.047
Method: probit	0.003	0.010	0.134	0.002	0.007	0.089
Method: tobit	0.007	0.016	0.181	0.003	0.010	0.085
Method: fixed-effects	0.001	0.005	0.035	0.000	0.003	0.022
Method: random-effects	-0.001	0.006	0.033	-0.002	0.007	0.052
Method: DID	0.000	0.003	0.016	0.000	0.002	0.013
Cross section	-0.029	0.008	0.995	-0.042	0.007	1.000
Publication characteristics						
Preferred estimate	0.007	0.009	0.398	0.001	0.003	0.072
Journal impact	0.002	0.006	0.159	0.002	0.005	0.148
Study citations	0.000	0.001	0.022	0.000	0.001	0.011
Observations	1,252			1,252		

Notes: The response variable is the partial correlation coefficient corresponding to the effect of financial incentives on performance reported in individual studies. SE = standard error, P. mean = posterior mean, P. SD = posterior standard deviation, PIP = posterior inclusion probability. The posterior mean in Bayesian model averaging denotes the marginal effect of a study characteristic on the partial correlation coefficient. The BMA specification in the left-hand portion of the table uses the BRIC g-prior based on Fernandez *et al.* (2001) and the beta-binominal model prior according to Ley & Steel (2009). The BMA specification in the right-hand portion of the table uses the same priors as the baseline BMA specification in the main text but additionally weights observations by inverse variance and the inverse of the number of estimates reported per study.

References

van Aert, R. C. & M. van Assen (2025): "Correcting for publication bias in a meta-analysis with the p-uniform* method." *Psychonomic Bulletin & Review* **(forthcoming)**.

Andrews, I. & M. Kasy (2019): "Identification of and correction for publication bias." *American Economic Review* **109(8)**: pp. 2766–94.

Angrist, J., D. Lang, & P. Oreopoulos (2009): "Incentives and services for college achievement: Evidence from a randomized trial." *American Economic Journal: Applied Economics* **1(1)**: pp. 136–63.

Ariely, D., A. Bracha, & S. Meier (2009): "Doing Good or Doing Well? Image Motivation and Monetary Incentives in Behaving Prosocially." *American Economic Review* **99(1)**: pp. 545–555.

Benabou, R. & J. Tirole (2003): "Intrinsic and Extrinsic Motivation." *The Review of Economic Studies* **70(3)**: pp. 489–520.

Benabou, R. & J. Tirole (2011): "Identity, Morals, and Taboos: Beliefs as Assets." *The Quarterly Journal of Economics* **126(2)**: pp. 805–855.

Bom, P. R. D. & H. Rachinger (2019): "A kinked meta-regression model for publication bias correction." *Research Synthesis Methods* **10(4)**: pp. 497–514.

Brown, A. L., T. Imai, F. Vieider, & C. Camerer (2024): "Meta-Analysis of Empirical Estimates of Loss-Aversion." *Journal of Economic Literature* **62(2)**: pp. 485–516.

Carrillo, P. E., E. Castro, & C. Scartascini (2021): "Public Good Provision and Property Tax Compliance: Evidence from a Natural Experiment." *Journal of Public Economics* **198**: p. 104422.

Charness, G. & U. Gneezy (2009): "Incentives to exercise." *Econometrica* **77(3)**: pp. 909–931.

Cox, G. & X. Shi (2023): "Simple adaptive size-exact testing for full-vector and subvector inference in moment inequality models." *The Review of Economic Studies* **90(1)**: pp. 201–228.

Dohmen, T. & A. Falk (2011): "Performance pay and multidimensional sorting: Productivity, preferences, and gender." *American Economic Review* **101(2)**: pp. 556–90.

Duflo, E., R. Hanna, & S. P. Ryan (2012): "Incentives work: Getting teachers to come to school." *American Economic Review* **102(4)**: pp. 1241–78.

Dwenger, N., H. Kleven, I. Rasul, & J. Rincke (2016): "Extrinsic and intrinsic motivations for tax compliance: Evidence from a field experiment in Germany." *American Economic Journal: Economic Policy* **8(3)**: pp. 203–32.

Elliott, G., N. Kudrin, & K. Wuthrich (2022): "Detecting p-hacking." *Econometrica* **90(2)**: pp. 887–906.

Elliott, G., N. Kudrin, & K. Wuthrich (2024): "The Power of Tests for Detecting p-Hacking." *working paper*, University of California, San Diego.

Englmaier, F., S. Grimm, D. Grothe, D. Schindler, & S. Schudy (2024): "The Effect of Incentives in Nonroutine Analytical Team Tasks." *Journal of Political Economy* **132(8)**: pp. 2695–2747.

Fernandez, C., E. Ley, & M. F. J. Steel (2001): "Benchmark priors for Bayesian Model Averaging." *Journal of Econometrics* **100(2)**: pp. 381–427.

Fershtman, C. & U. Gneezy (2011): "The trade-off between performance and quitting in high power tournaments." *Journal of the European Economic Association* **9(2)**: pp. 318–336.

Frey, B. S. & F. Oberholzer-Gee (1997): "The Cost of Price Incentives: An Empirical Analysis of Motivation Crowding-Out." *American Economic Review* **87(4)**: pp. 746–755.

Furukawa, C. (2020): "Publication bias under aggregation frictions: Theory, evidence, and a new correction method." *Working paper*, MIT.

Gerber, A. & N. Malhotra (2008): "Do Statistical Reporting Standards Affect What Is Published? Publication Bias in Two Leading Political Science Journals." *Quarterly Journal of Political Science* **3(3)**: pp. 313–326.

Gneezy, U. & J. A. List (2006): "Putting Behavioral Economics to Work: Testing for Gift Exchange in Labor Markets Using Field Experiments." *Econometrica* **74(5)**: pp. 1365–1384.

Gneezy, U. & A. Rustichini (2000): "Pay enough or don't pay at all." *The Quarterly Journal of Economics* **115(3)**: pp. 791–810.

Havranek, T., Z. Irsova, L. Laslopova, & O. Zeynalova (2024): "Publication and Attenuation Biases in Measuring Skill Substitution." *The Review of Economics and Statistics* **106(5)**: pp. 1187–1200.

Ioannidis, J. P., T. D. Stanley, & H. Doucouliagos (2017): "The Power of Bias in Economics Research." *Economic Journal* **127(605)**: pp. F236–F265.

Irsova, Z., P. R. D. Bom, T. Havranek, & H. Rachinger (2025): "Spurious Precision in Meta-Analysis of Observational Research." *Nature Communications* **16**: p. 8454.

Ito, K. (2015): "Asymmetric Incentives in Subsidies: Evidence from a Large-Scale Electricity Rebate Program." *American Economic Journal: Economic Policy* **7(3)**: pp. 209–237.

Karlan, D. & J. A. List (2007): "Does price matter in charitable giving? Evidence from a large-scale natural field experiment." *American Economic Review* **97(5)**: pp. 1774–1793.

Kim, J. H., B. Gerhart, & M. Fang (2022): "Do

Financial Incentives Help or Harm Performance in Interesting Tasks?" *Journal of Applied Psychology* **107(1)**: pp. 153–167.

Kranz, S. & P. Putz (2022): "Methods Matter: p-Hacking and Publication Bias in Causal Analysis in Economics: Comment." *American Economic Review* **112(9)**: pp. 3124–36.

Lazear, E. P. (2000): "Performance pay and productivity." *American Economic Review* **90(5)**: pp. 1346–1361.

Levitt, S. D., J. A. List, S. Neckermann, & S. Sadoff (2016): "The behavioralist goes to school: Leveraging behavioral economics to improve educational performance." *American Economic Journal: Economic Policy* **8(4)**: pp. 183–219.

Ley, E. & M. F. Steel (2009): "On the Effect of Prior Assumptions in Bayesian Model Averaging with Applications to Growth Regression." *Journal of Applied Econometrics* **24(4)**: pp. 651–674.

Matousek, J., T. Havranek, & Z. Irsova (2022): "Individual Discount Rates: A Meta-Analysis of Experimental Evidence." *Experimental Economics* **25(1)**: pp. 318–358.

Meier, S. (2007): "Do subsidies increase charitable giving in the long run? Matching donations in a field experiment." *Journal of the European Economic Association* **5(6)**: pp. 1203–1222.

Sliwka, D. (2007): "Trust as a Signal of a Social Norm and the Hidden Costs of Incentive Schemes." *American Economic Review* **97(3)**: pp. 999–1012.

Sliwka, D. & P. Werner (2017): "Wage increases and the dynamics of reciprocity." *Journal of Labor Economics* **35(2)**: pp. 299–344.

Stanley, T. D., S. B. Jarrell, & H. Doucouliagos (2010): "Could it be better to discard 90% of the data? A statistical paradox." *The American Statistician* **64(1)**: pp. 70–77.

Tirole, J. & R. Benabou (2006): "Incentives and Prosocial Behavior." *American Economic Review* **96(5)**: pp. 1652–1678.

Weibel, A., K. Rost, & M. Osterloh (2010): "Pay for Performance in the Public Sector—Benefits and (Hidden) Costs." *Journal of Public Administration Research and Theory* **20(2)**: pp. 387–412.